Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21313

F-413

B.C.A. (Third Semester)

Examination, 2024-25

(New Course)

Paper - BCA-303-N

Data Structure Using C

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks i.e. 14 marks symbols used have their usual meanings.

1. (a) Explain the basic terminology of data structures. What are the different types of data organizations used in programming?

   (b) Discuss the concept of algorithm complexity and time-space trade-off with examples.

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2. (a) Define arrays. Explain the representation and analysis of single and multidimensional arrays. How is the address of an element calculated in a one -dimensional array?

   (b) Write a C program to perform basic operations (insertion, deletion) on a one-dimensional array.

3. (a) Construct an expression tree for the expression (a+b*c)+((d*e+f)*g). Give the outputs when you apply inorder, preorder and postorder traversals.

   (b) Create a binary search tree for the following numbers start from an empty binary search tree. 45, 26, 10, 60, 70, 30, 40 Delete keys 10, 60 and 45 one after the other and show the trees at each stage.

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4. (a) How does quick sort algorithm works? What is the output of quick sort after the 3rd iteration given the following sequence?
24 56 47 35 10 90 82 31

(b) What are the two traversal strategies used in traversing a graph?

5. (a) Write an algorithm for Push and Pop operations on Stack using Linked list.

(b) What is a DeQueue? Explain its operation with example? https://www.mjpruonline.com

6. (a) Define a graph. Explain the different types of graphs and their representations.

(b) Describe the adjacency matrix representation of a graph. How is it used for graph traversal?

7. (a) Explain the concept of hashing. Describe different collision resolution strategies used in hash tables.

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(b) Show the result of inserting the keys 2,3,5,7,11,13,15,6,4 into an initially empty extendible hashing data structure with M=3.

8. Write short notes on any **four** of the following:

(a) Biconnected Graph

(b) Sparse Matrix

(c) Vector

(d) Two-way Header List

(e) Recursion

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